higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 4

15 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 True
1.2 True
1.3 False
1.4 False
1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1 G
2.2 F
2.3 I
2.4 B
2.5 E

(5 × 1) **[5]**

QUESTION 3

3.1 3.1.1 Data derived from discrete measurement such as true or false, percentage of functional elements, number of defective elements.

OR

Data represented in discrete units like hours, minutes or seconds which measure the presence or absence of characteristics of a responsive variable in the form of bar charts, control charts, etc.

3.1.2 Bar graph which shows how frequently data falls within specific cells or a graph that estimates where the process is centred and how much fluctuation there is from the centre point.

3.1.3 A process for detecting potential problems or non-conformances and eliminating them.

(3 × 2) (6)

3.2
- Products must meet standards specified by law.
- Products must meet standards set by SABS.
- Products must meet standards set by various international bodies.
- Products must satisfy the market requirements or standards set by consumers. (4)

3.3
- An X-chart monitor represents quantitative characteristics of processes.
- An R-chart represents the range of the variable being measured, i.e. the

difference between the highest variable and the lowest variable being measured. (2 × 2) (4)

3.4
- Reviewing and defining the problem or non-conformity.
- Finding the cause of the problem.
- Developing an action plan to correct the problem and prevent recurrence.
- Implementing the plan.
- Evaluating the effectiveness of the correction. (5)

3.5 Corrective Action/Preventative Action. (1)

3.6
- Absolute pressure:✓ It is the difference in pressure between the point of measurement and a perfect vacuum where the pressure is zero.✓
- Gauge pressure:✓ It is the pressure difference between the point of measurement and ambient or atmospheric pressure.✓
- Differential pressure:✓ It is the pressure difference between any two points with one chosen as reference.✓ (3 × 2) (6)

3.7 °R = °F + 459,6✓
= 149 + 459,6✓
= 608,6✓✓ (4)

[30]

QUESTION 4

4.1 They are used as powerful tools for process control. (2)

4.2
- SCADA is data gathering-oriented.
- DCS is process-oriented. (2 × 2) (4)

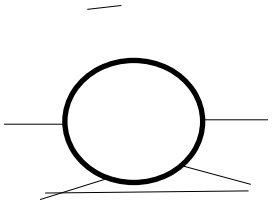
4.3
- Gas chromatography is used to measure stream composition.
- Thermocouples and resistance thermometers are used to measure temperature.
- Venturi and orifice plates are used to measure stream flow rate. (3 × 2) (6)

4.4
- Show the process flow in the plant.
- Show the actual up to date process conditions in the plant.
- Give warning signals where alarm conditions are encountered.
- Allow changing of set points where required.
- Retrieve the historical data to monitor trends and recall information.

(Any 3 × 1) (3)

4.5 4.5.1

4.5.2



(2 × 2) (4)

4.6
- To check settings of a previous batch.
- Find out why deviation occurred.
- Find out why a product's quality is not good.
- Find out why an accident occurred.

OR

- Problem solving.
- Comparison or selection of running conditions.
- Trend analysis.
- Best practice determination.
- Training. (4)

4.7 They are small acceptable deviations that occur naturally in all processes. (2)
[25]

QUESTION 5

5.1
- Identify all the symptoms.
- Define all the possible causes.
- Eliminate some of the possible cause.
- Identify the probable cause.
- Define possible solutions.
- Select and implement the best solution. (6)

5.2 To give guidance to bring a process online starting from a non-operational system. (2)

5.3
- Type and size of the plant.
- The processes.
- Chemicals and the type of unit operations involved.
- Type of machinery involved.
- The location.
- Type of utilities. (5)

5.4 Alarms indicate a problem or potential problem. (2)

5.5
- Product.
- Pressure.
- Level.
- Flow.
- Temperature.
- Electrical and mechanical abnormalities. (5)

[20]

QUESTION 6

6.1
- Manual control.
- Automatic control.
- Cascade control. (3)

6.2 Programmable Logic Control. (1)

6.3
- Sequencing.
- Interlock operations.
- Systems that contain equipment that operates discontinuously.
- Non-continuous systems such as batch processes. (4)

6.4
- Single action.
- Repetitive action.
- Once-off applications with complex manipulations. (3)

6.5
- Adders.
- Multipliers.
- Low limiters.
- High limiters. (4)

[15]

TOTAL: 100

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